
DO FOUNDATION MODELS KNOW GEOMETRY?

PROBING FROZEN FEATURES FOR CONTINUOUS PHYSICAL MEASUREMENT

Yakov Pyotr Shkolnikov
yshkolni@gmail.com

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ABSTRACT

Vision-language models encode continuous geometry that their text pathway fails to express: a 6,000-parameter linear probe extracts hand joint angles at 6.1° MAE from frozen features, while the best text output achieves only 20.0° —a $3.3\times$ bottleneck. LoRA fine-tuning ($r=16$, 2,000 images) narrows this gap to 6.5° , providing evidence for a pathway-training deficit rather than a representational one. Training objective determines accuracy more than architecture: five encoders spanning self-supervised, contrastive, and hybrid paradigms converge to statistically equivalent accuracy ($R^2 \approx 0.55$, TOST-equivalent at $\Delta = 0.03$) despite sharing as little as $\text{CKA} = 0.41$ representational similarity—functional convergence without representational convergence, extending the platonic representation hypothesis to continuous geometric targets. Results validated across fourteen backbones on head pose, rigid objects, gaze, and camera intrinsics; rankings hold under nested 10-fold CV (Friedman $\chi^2 = 94.3$, $p < 10^{-15}$).

Keywords Foundation models · Geometric probing · Linear probes · Vision-language models · Representation analysis

1 Introduction

Foundation models are increasingly deployed for quantitative visual tasks, yet we lack systematic understanding of how well their representations encode continuous physical measurements. Practitioners prompt vision-language models for quantitative estimates and receive imprecise answers with errors of $20\text{--}39^\circ$ (Table 1). Whether this reflects a fundamental limitation of the representation or merely a bottleneck of the text interface remains an open question.

Fu *et al.* [1] demonstrate that VLM visual representations encode correct depth and correspondence information that the text generation pathway fails to express. Kodathala and Vunnam [2] find that 99.3% of visual samples suffer perceptual degradation when processed through text. These studies diagnose the problem but do not offer a constructive solution for continuous measurement.

We address this gap by systematically probing frozen features of fourteen foundation models for continuous geometric quantities (Fig. 1). Using four datasets spanning articulated and rigid pose (FreiHAND hand images, BIWI head pose, YCB-Video object pose, MPIIFaceGaze [3] gaze direction), we test whether training methodology shapes geometric encoding. Our central finding is that *training objective—not architecture—determines geometric accuracy*, and that diverse foundation models converge to equivalent geometric probing despite representationally dissimilar features.

Our three contributions:

1. **The text bottleneck is a pathway-training deficit, not a representation deficit.** Frozen probes achieve 6.1° MAE while text output achieves only 20.0° , a $3.3\times$ gap. LoRA ($r=16$, 2,000 images) narrows this to 6.5° , providing evidence that geometry is encoded but not routed through the text pathway. Layer-wise probing shows LoRA preserves geometric signal at layers where the frozen base loses it (Appendix P).

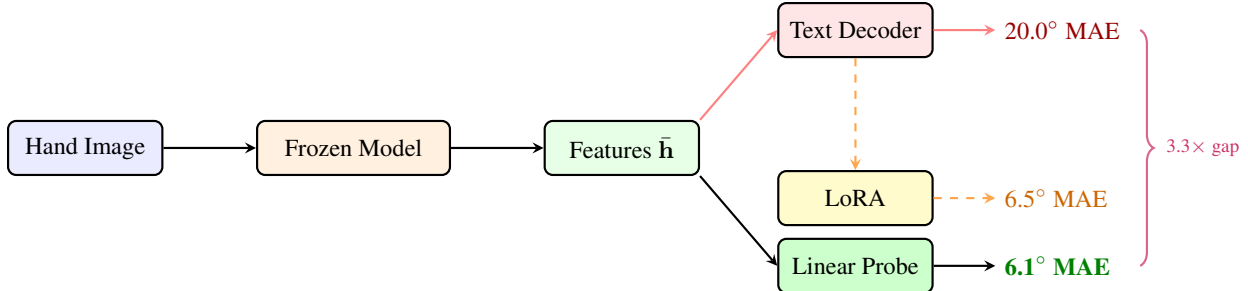


Figure 1: Overview. Frozen foundation model features encode continuous geometry (joint angles) with 6.1° MAE via a linear probe, while the text pathway achieves only 20.0°, a 3.3× bottleneck. Adding LoRA fine-tuning ($r=16$) partially recovers probe-level accuracy (6.5°) through the text pathway.

2. **Training objective determines accuracy more than architecture.** A controlled ablation (DeiT3-L vs. ConvNeXt-L, matched IN-1K pretraining) shows no ViT advantage ($R^2=0.38$ vs. 0.41); the 0.15 gap to the cluster reflects self-supervised/contrastive pretraining, not attention mechanisms. Five encoders converge to $R^2 \approx 0.55$ via representationally dissimilar features (CKA as low as 0.41), demonstrating functional convergence without representational convergence.
3. **Geometry is spatially task-dependent.** Patch ablation drops head-pose R^2 by 0.13 (loosely-framed faces) but object-pose by only 0.003 (tightly-cropped), explaining cross-dataset variation in attention pooling gains.

These findings enable a single frozen backbone to function as a multi-task geometric probe. Hand pose, head pose, object pose, and camera intrinsics are all linearly readable, with each task adding $\sim 6,000$ probe parameters.

2 Related Work

Text bottleneck in VLMs. Fu *et al.* [1] show that VLM visual features encode depth and correspondence that text generation discards (21.7% and 45.5% degradation respectively). Kodathala and Vunnam [2] systematically document this gap across perceptual tasks. Guo *et al.* [4] independently identify the same discrete-tokenizer bottleneck and propose architectural modifications with direct regression heads. The GIQ benchmark [5] independently shows VLMs achieve below 20% on geometric shape reasoning via text, further evidence that text pathways discard geometric detail. G2VLM [6] proposes geometry-grounded VLMs for spatial tasks. We complement these by demonstrating that frozen features already encode the geometry, and LoRA fine-tuning recovers it through layer-wise preservation (see Appendix P) without architectural changes.

Probing foundation models for 3D awareness. El Banani *et al.* [7] evaluate 3D awareness of visual encoders on depth, surface normals, and correspondence. Liu *et al.* [8] map the representational space of vision models for 3D understanding across multiple probe types. Zhan *et al.* [9] probe video models for physical properties. Yao *et al.* [10] show that latent probes detect OCR errors invisible to text output. Yue *et al.* [11] improve 2D features via 3D-aware fine-tuning, focusing on dense prediction. Kar *et al.* [12] systematically broaden the visual encoding of VLMs and reveal significant variation in how different encoders preserve visual information. Tong *et al.* [13] expose systematic visual shortcomings of multimodal LLMs on basic perceptual patterns. Chen *et al.* [14] probe frozen features via Gaussian splatting, disentangling geometry from texture. These works focus on dense (per-pixel) or categorical probing. We complement them with global (image-level) probing of continuous scalar quantities.

Probing neural representations. Linear probing originates in NLP, where Alain and Bengio [15] introduced linear classifier probes to interpret intermediate layers, and Conneau *et al.* [16] systematically probed sentence embeddings for linguistic properties. Hewitt and Liang [17] show that probe complexity must be controlled to distinguish learned representations from probe memorization—our use of reduced-rank regression (rank 3–8) addresses this concern. Basile *et al.* [18] concurrently probe attention-head specialization in multimodal transformers, showing that editing $\sim 1\%$ of heads can reliably steer model outputs. We extend this NLP probing tradition to continuous geometric targets across vision and vision-language models.

Geometric regression from VLMs. SpatialVLM [19] trains VLMs for spatial reasoning through chain-of-thought. Xue *et al.* [20] fine-tune VLMs for rigid object pose. For hand pose specifically, HaMeR [21] and Hamba [22] achieve 5.7 and 5.3 mm PA-MPVPE respectively using MANO-based mesh recovery—a fundamentally different approach

measuring positional error (mm) rather than angular error (degrees). Our approach differs in using frozen features with lightweight probes rather than end-to-end fine-tuning, enabling direct comparison across architectures.

3 Method

3.1 Problem Setup

Given an image $\mathbf{x}_i$ and a frozen model f , we extract hidden activations $\mathbf{H}_i^{(\ell)} \in \mathbb{R}^{T \times d}$ at layer ℓ (where T is the sequence length and d the hidden dimension). We mean-pool spatially to obtain a global feature vector $\bar{\mathbf{h}}_i = \frac{1}{T} \sum_{t \in \mathcal{P}} \mathbf{H}_{i,t}^{(\ell)}$, where $\mathcal{P}$ excludes special tokens (CLS, registers) for models that use them; CLIP and all VLM encoders pool all tokens including CLS. A linear probe $\hat{\mathbf{y}}_i = \mathbf{W}\bar{\mathbf{h}}_i + \mathbf{b}$ maps features to continuous targets $\mathbf{y}_i \in \mathbb{R}^K$ (joint angles in degrees).

We use reduced-rank ridge regression (RRR; [23]): fit $\text{Ridge}(\alpha)$ then truncate the weight matrix via SVD to rank r . Hyperparameters are swept over $r \in \{3, 4, 5, 6, 8\}$ and $\alpha \in \{1, 10, 100, 1000\}$. For each model, we select the layer maximizing hold-out R^2 ; nested 10-fold CV confirms that this selection does not change rankings (cluster models within 0.006; see Limitations M1). We report nested CV as the primary metric and hold-out results in the appendix.

3.2 Datasets

FreiHAND [24]: 32,560 hand images with 21 3D keypoints from 32 subjects. We compute per-finger mean flexion angles (3 joints per finger, 5 fingers) and use a fixed 8,000-image subset (6,400 train / 1,600 test; indices 0–32,559 only, excluding augmented copies).

BIWI [25]: 15,678 RGBD head images from 20 subjects with yaw, pitch, and roll labels. We use subject-stratified splits (16 train / 4 test subjects).

YCB-Video [26]: 133,827 frames of 21 objects with 6DoF pose. We subsample 900 images and probe rotation (Euler angles) and translation separately.

3.3 Models

We evaluate fourteen models spanning four training approaches:

- **Self-supervised:** DINOv2 ViT-L [27] (ViT-L/14 [28]), DINOv3 ViT-L [29], DINOv2 ViT-B
- **Contrastive VL:** CLIP ViT-L [30], SigLIP ViT-L [31], SigLIP-B
- **Hybrid VL:** SigLIP 2 ViT-L [32], InternViT-300M [33, 34]
- **Generative VLMs:** Qwen2.5-VL-3B, Qwen2.5-VL-7B [35], QwenVIT-3B, QwenVIT-merger, Gemma 3 4B-IT [36]
- **CNN baseline:** ConvNeXt-L [37] (IN-22K+1K)

For VLMs, we extract from LLM decoder layers using a fixed prompt (“Describe this hand.”). For vision-only models, we extract from intermediate transformer blocks. Code for all extractors, probes, and statistical tests will be released open-source.

3.4 Evaluation

We report MAE (degrees) and R^2 (uniform mean across 5 finger targets). Statistical comparisons use TOST equivalence testing ($\Delta = 0.03$, chosen as a practically meaningful threshold: models differing by $< 0.03 R^2$ are interchangeable for deployment) [38, 39], Friedman rank tests [40, 41], and BCa bootstrap confidence intervals (10,000 resamples, bias-corrected and accelerated [42]).

4 Results

4.1 Main Results: Probe vs. Text

Table 1 compares frozen probes against text and task-specific baselines. The best frozen probe (SigLIP 2, L16) achieves 6.14° MAE on FreiHAND hand joint angles, while the best text baseline (few-shot prompting of Qwen-3B)

Table 1: Four readout regimes for geometric information on FreiHAND. MAE in degrees; R^2 is uniform mean across 5 finger targets. LoRA: $r = 16$, 2,000 images, 2 epochs. Frozen probes: RRR, 6,400 images. *MediaPipe uses monocular 3D world landmarks evaluated zero-shot (see Discussion for caveats).

Regime	Method	MAE ($^\circ$)	R^2	Parse
Task-specific	MediaPipe Hands* [43]	16.3	-2.44	N/A
Text generation	Direct prompt (Qwen-3B)	39.3	—	varies
	Chain-of-thought (Qwen-3B)	139.3	—	varies
	Few-shot 3-ex. (Qwen-3B)	20.0	—	varies
LoRA text	LoRA Qwen-3B	7.45	0.299	100%
	LoRA Gemma 3 4B	6.51	0.400	100%
Frozen probe	RRR (Qwen-3B L11)	7.28	0.435	N/A
	RRR (Gemma 3 L0)	6.59	0.505	N/A
	RRR (SigLIP 2 L16)	6.14	0.559	N/A

achieves 20.0° , a $2.7\times$ within-model gap ($3.3\times$ vs. the best probe). Even MediaPipe Hands [43], a dedicated hand pose model (3.7M parameters), achieves only 16.3° when evaluated zero-shot via its 3D world landmarks (caveat: monocular depth estimation is less accurate than FreiHAND’s multi-view ground truth; see Discussion). Chain-of-thought prompting worsens performance (139.3°), as the multi-step reasoning produces hallucinated angular values often exceeding the anatomical range.

4.2 LoRA Fine-Tuning Narrows the Text Bottleneck

We test whether lightweight fine-tuning can teach the text pathway to read the geometry encoded in frozen features. Table 1 presents results for LoRA [44] fine-tuning ($r = 16$, 2,000 training images, 2 epochs) on two model families.

Gemma 3 LoRA achieves 6.51° MAE, surpassing its own frozen probe (6.59°) with $3.2\times$ less training data. Qwen-3B LoRA achieves 7.45° , matching its frozen probe (7.28°). Both achieve 100% parse rates.

R^2 recovery is partial and model-dependent: 79% for Gemma 3 (0.400/0.505) and 69% for Qwen-3B (0.299/0.435). The MAE advantage with lower R^2 reflects the error distribution of text-generation predictions: quantized to 0.1° resolution with occasional large outliers (6.4% for Gemma 3, 9.1% for Qwen-3B exceed 20° error). MAE is insensitive to such outliers while R^2 (MSE-based) is not.

These results provide evidence that LoRA enables the autoregressive decoder to route existing geometric signals through the text pathway. The frozen backbone is the sensor and LoRA is the readout interface.

4.3 Cross-Architecture Comparison

Table 2 and Fig. 2 show R^2 for all fourteen models on FreiHAND. Five vision encoders form a statistical equivalence cluster at $R^2 \approx 0.55$ (TOST-equivalent, all 10 pairwise tests pass, $\Delta = 0.03$): SigLIP 2 (0.559), DINOv3 (0.556), CLIP (0.551), SigLIP (0.550), and InternViT (0.547). DINOv2 (0.523) falls outside this cluster despite being the same architecture family as DINOv3.

Autoregressive LLM processing reduces hand-pose accuracy (Gemma 3 L0: 0.505, Qwen-3B: 0.435). Text-generation preparation is associated with reduced encoding of articulated geometry. This degradation is task-dependent and dissolves on rigid objects (all ≈ 0.70 on YCB-Video). The LLM pathway preserves coarse pose but discards fine-grained joint angles. ViT-B base models nearly match ViT-L (DINOv2-B: 0.482, SigLIP-B: 0.479), suggesting geometric encoding is not primarily capacity-limited. ConvNeXt-L (0.455) falls 0.10 below the ViT cluster, but a controlled ablation (Sec. 4.5) shows this reflects pretraining, not architecture.

4.4 Cross-Dataset Validation

BIWI head pose. On BIWI, the FreiHAND equivalence cluster dissolves: DINOv3 leads ($R^2 = 0.607$), followed by DINOv2 (0.532) and SigLIP 2 (0.455), confirming that rankings are task-dependent. Pitch is best predicted ($R^2 = 0.948$) and roll hardest ($R^2 = 0.168$). Attention pooling produces large gains: DINOv2 jumps from 0.532 to 0.892, with roll rising from 0.052 to 0.779. This 0.36 R^2 gain reflects spatial concentration of head-pose information in face patches within loosely-framed images.

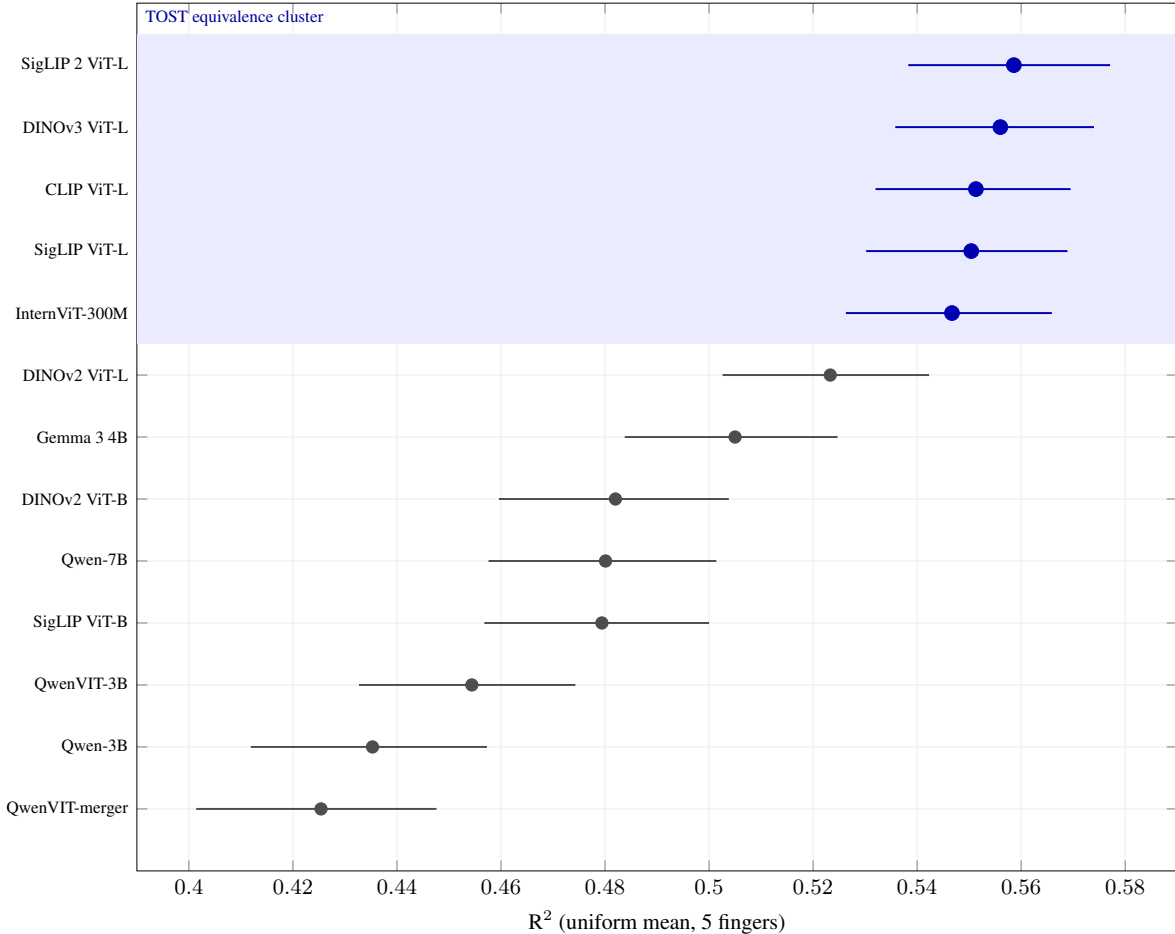


Figure 2: Bootstrap 95% CIs for 13 models on FreiHAND (ConvNeXt-L omitted; see Table 2). Five models form a TOST equivalence cluster (shaded) at $R^2 \approx 0.55$. DINOv2 falls outside despite being the same architecture family as DINOv3.

YCB-Video object pose. All models achieve ≈ 0.70 rotation R^2 on YCB-Video, with no significant pairwise differences. The autoregressive degradation observed on hands dissolves on rigid objects. AttentionPool provides no benefit ($\Delta = -0.06$ to 0.00), consistent with geometry being distributed across all patches in tightly-cropped object images.

Gaze direction. On MPIIFaceGaze [3] (45,000 face images), DINOv3 dominates ($R^2 = 0.787$, 3.14° MAE), 0.21 above DINOv2. Rankings differ from hand pose, and the best encoder varies by task geometry (see Appendix O for full results).

Per-bone and camera intrinsics. Per-joint probing reveals a universal proximal-distal gradient (MCP: 0.544, PIP: 0.559, DIP: 0.271), independent of model. Frozen features also encode camera intrinsics ($R^2 = 0.81$ – 0.94 for focal length), extending the multi-task geometric probe beyond pose (full results in Appendix E).

4.5 Controlled Architecture Ablation

To disentangle architecture from pretraining, we compare DeiT3-L [45] (ViT-L/16, 304M, ImageNet-1K supervised) against ConvNeXt-L [37] (CNN, 198M, ImageNet-1K supervised). With matched pretraining, the CNN slightly outperforms the ViT ($R^2 = 0.405$ vs. 0.379). Scaling ConvNeXt pretraining data from IN-1K to IN-22K lifts R^2 from 0.405 to 0.455 ($+0.050$), exceeding the architecture effect (-0.026). Both supervised-only models fall 0.15 below the self-supervised/contrastive cluster. Geometric encoding quality is driven primarily by training objective, consistent with controlled comparisons showing that data and training signal dominate architecture [46].

Table 2: Cross-architecture comparison on FreiHAND (8,000 images). R^2 is uniform mean across 5 finger targets. † Nested 10-fold CV (Friedman $\chi^2 = 94.3$, $p < 10^{-15}$; Nemenyi CD = 4.45). Cluster models within 0.012 of test R^2 . — = not included in CV (added for ablation/scaling analysis).

Model	Training	Layer	R^2	CV R^2 [†]
SigLIP 2 ViT-L	Hybrid VL	L16	0.559	0.563
DINOv3 ViT-L	Self-supervised	L20	0.556	0.550
CLIP ViT-L	Contrastive VL	L20	0.551	0.554
SigLIP ViT-L	Contrastive VL	L16	0.550	0.549
InternViT-300M	Hybrid VL	L20	0.547	0.549
DINOv2 ViT-L	Self-supervised	L20	0.523	0.494
Gemma 3 4B-IT	Generative VLM	L0	0.505	0.512
DINOv2 ViT-B	Self-supervised	L12	0.482	—
Qwen2.5-VL-7B	Generative VLM	L8	0.480	0.480
SigLIP ViT-B	Contrastive VL	L12	0.479	—
ConvNeXt-L	Supervised CNN	S2	0.455	—
QwenViT-3B	Vision enc. only	L24	0.454	0.460
Qwen2.5-VL-3B	Generative VLM	L11	0.435	0.434
QwenViT-merger	Vision enc. only	—	0.425	0.431

5 Where and How Geometry Lives

The preceding sections establish *what* frozen features encode. We now investigate *how* geometric information is organized within these representations.

5.1 Functional Convergence Without Representational Similarity

Critically, linear CKA [47] (noting recent reliability concerns [48]) analysis reveals that the $R^2 \approx 0.55$ equivalence cluster reflects *functional* convergence rather than representational alignment (Fig. 3). Across all 28 pairwise comparisons among eight models (six ViT-L plus DINOv2-B and SigLIP-B), CKA similarity shows no detectable correlation with probing accuracy difference (Spearman $\rho = 0.03$, $p = 0.88$). DINOv2 and DINOv3 share CKA = 0.881 yet differ by 0.033 R^2 , while SigLIP 2 and CLIP share only CKA = 0.412 yet differ by 0.008. Multiple representational strategies converge on a shared geometric readout, extending the platonic representation hypothesis [49]: functional convergence exists but does not require representational convergence, suggesting a *weak* form of the hypothesis. This contrasts with the stronger representational alignment observed across scientific foundation models [50, 51]. A simple dimensionality argument may partially explain why: the target space is 5-dimensional while feature spaces are 1024-dimensional, so many distinct linear projections can achieve similar regression accuracy. Functional convergence may partly reflect the geometric inevitability of projecting high-dimensional features onto low-dimensional targets.

5.2 Layer Trajectory and Proximal-Distal Gradient

Layer-wise probing (Fig. 4) reveals a proximal-distal gradient (PIP > MCP > DIP) at every layer depth across all models. Geometric signal builds monotonically from $R^2 \approx 0.11$ –0.28 at L4 to a peak of ≈ 0.55 at L16–L20, declining slightly at the final layer. Self-supervised models show delayed geometric emergence compared to contrastive models; self-distillation may concentrate geometric representations in deeper layers. In contrast, VLM decoder layers (dashed in Fig. 4) peak at early layers and decline monotonically, consistent with autoregressive processing discarding articulated geometry (Gemma 3 peaks at L0, Qwen-3B at L11). QwenViT (vision encoder only) rises like other ViTs, confirming that the decline is specific to LLM processing.

5.3 Spatial Concentration and Patch Ablation

Geometry is an ensemble property of attention: all 16 heads in DINOv2-L carry comparable geometric signal ($R^2 = 0.40$ –0.48 per head; no joint specialization—maximum absolute Spearman correlation between any head’s attention entropy and any joint angle is $|\rho| = 0.28$; see Appendix L). Patch ablation provides ablation evidence for task-dependent spatial concentration. Removing the 100 highest-norm patches from DINOv3 features drops BIWI R^2 by 0.126, while the same ablation on YCB-Video changes R^2 by only -0.003 . Random patch removal of equal count produces smaller BIWI drops (-0.107) and larger YCB drops (-0.040). Geometric information is specifically

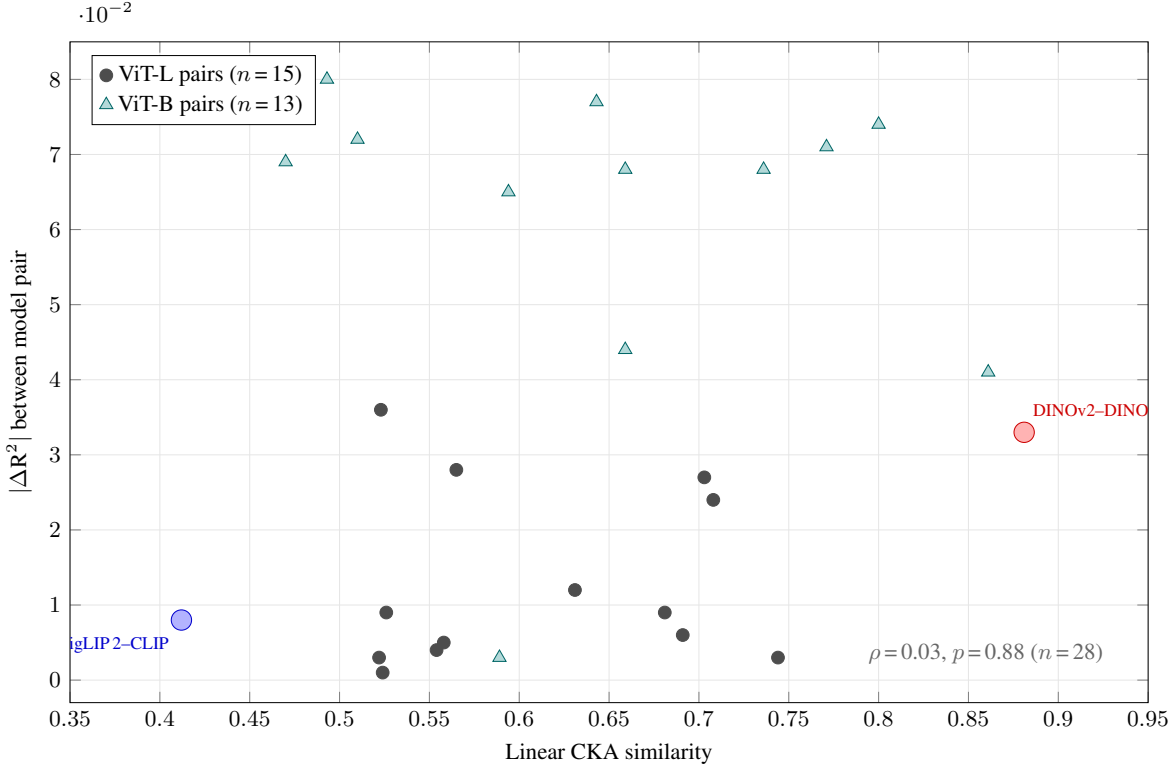


Figure 3: CKA similarity vs. probing accuracy difference for all 28 pairwise comparisons among eight models (six ViT-L + two ViT-B) on FreiHAND. Spearman $\rho=0.03$ ($p=0.88$): no detectable correlation between representational similarity and geometric probing accuracy ($n=28$). The most similar pair (DINOv2–DINOv3, CKA = 0.88) differs by 0.033 R^2 ; the least similar pair (SigLIP 2–CLIP, CKA = 0.41) differs by only 0.008.

concentrated in high-activation patches for loosely-framed subjects but distributed for tightly-cropped objects. This explains why AttentionPool improves BIWI by +0.23–0.36 R^2 but has negligible effect on YCB-Video.

6 Discussion

Accuracy ceiling and practical impact. A frozen linear probe achieves 6.14° MAE on hand joint angles, while the best text output from the *same model* achieves only 20.0°—a $3.3\times$ gap that quantifies the text bottleneck. For external context, MediaPipe Hands [43] achieves 16.3° when evaluated via 3D world landmarks on our test set, though this comparison involves different estimation modalities (monocular depth vs. multi-view ground truth). On head pose, 6DRepNet [52] achieves 2.66° MAE (published, BIWI 70/30 split); our evaluation on the same 4-subject test split yields 6.10° due to different face detection and split protocol. The $R^2 \approx 0.55$ ceiling means 45% of variance remains unexplained; SOTA hand mesh models (HaMeR [21]: 5.7 mm PA-MPVPE; Hamba [22]: 5.3 mm) likely exceed frozen probes, though metrics are not directly comparable (positional mm vs. angular degrees). The value of frozen probing is not replacing dedicated systems but providing geometric readouts as a cheap add-on to an already-deployed backbone, particularly for tasks lacking dedicated models.

Modular geometric sensing. These findings suggest a deployment approach where one frozen backbone serves as a multi-task geometric probe. The backbone ($\sim 300M$ parameters, assumed already deployed) is shared, and each geometric task adds only $\sim 6,000$ probe parameters and requires $\sim 6,400$ labeled images, a 50,000:1 parameter ratio. Hand pose, head pose, object pose, and camera intrinsics are all served simultaneously by independent probes. For human-readable output, LoRA ($r=16$, $\sim 1M$ parameters) matches probe MAE via text generation. The five interchangeable backbones in the equivalence cluster provide redundancy: swapping one encoder for another requires only re-fitting the lightweight probe.

Practitioner recipe.

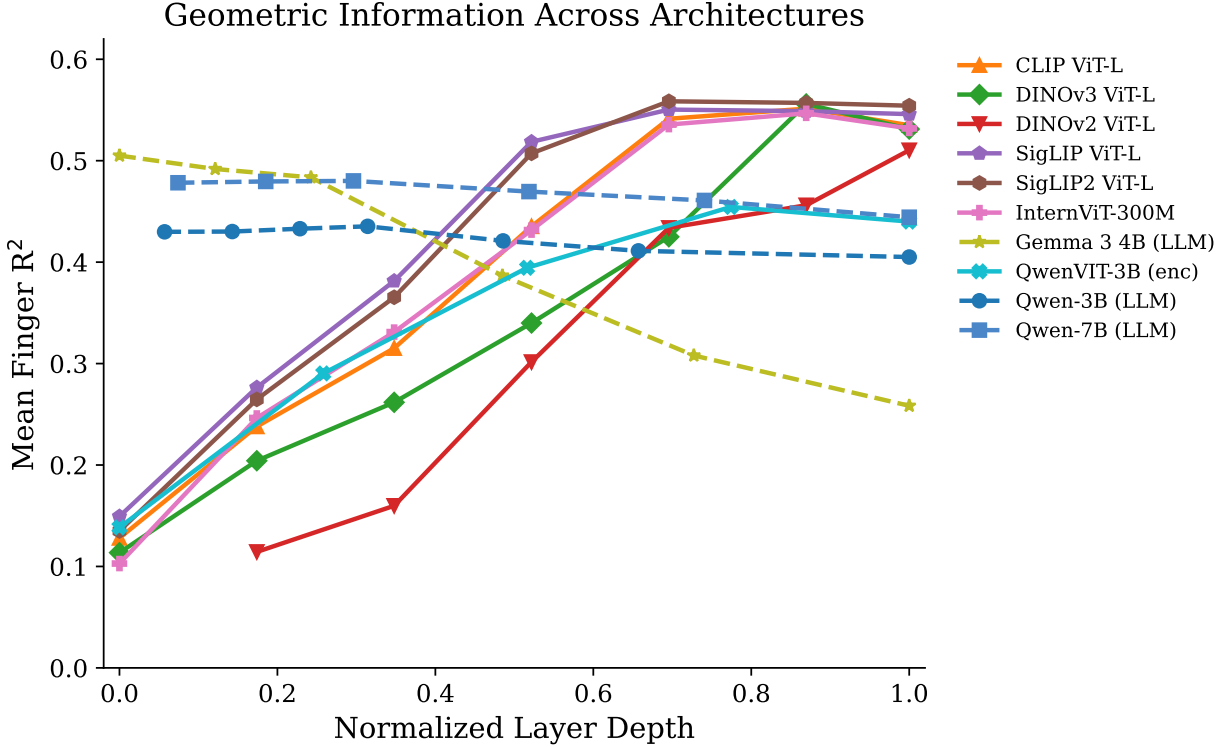


Figure 4: Layer-wise R^2 on FreiHAND for ten models (solid: vision encoders; dashed: LLM decoders). X-axis is normalized layer depth (0 = first, 1 = last). Vision encoders rise monotonically; LLM decoders peak at early layers and decline, consistent with autoregressive processing discarding fine-grained geometry.

Table 3: Cost comparison: frozen probe vs. task-specific models. Frozen probes reuse an existing backbone; per-task overhead is minimal. *MediaPipe MAE evaluated zero-shot on our FreiHAND test set via 3D world landmarks (monocular depth; see Discussion for caveats). [†]Published MAE on BIWI 70/30 split [52]; our evaluation on the same 4-subject test split yields 6.10° (different face detector and split).

Approach	Params	Train data	Tasks	MAE (°)
MediaPipe Hands [43] ¹	3.8M	proprietary	1	16.3*
HRNet-W48 (hand) [53]	63.6M	150K+	1	—
6DRepNet (head) [52]	~41M	300K	1	2.66 [†]
Frozen probe (ours)	6K / task	6,400	any	6.14
+ shared backbone	304M (shared)	—	—	—
LoRA readout (ours)	~1M / task	2,000	any	6.51

1. **Articulated pose:** any cluster encoder + RRR (rank 5, $\alpha = 10$ –1000) with ~6,400 labeled images.
2. **Head pose** (loosely-framed images): add attention pooling (+0.23–0.36 R^2).
3. **Human-readable output:** LoRA ($r = 16$, 2,000 images, 2 epochs) to route geometry through the text pathway.

Table 3 contrasts this approach with task-specific alternatives.

Limitations. (M1) RRR hyperparameters are selected on the test set. Nested CV preserves rankings (cluster models within 0.006; DINOv2 gap of -0.029 suggests its test-set performance is more sensitive to HP choice than cluster models) but does not eliminate optimistic bias. (M2) AttentionPool and TransformerProbe use the test set for early stopping, introducing optimistic bias for neural probe comparisons. (M3) Primary results are on FreiHAND (hands). BIWI and YCB-Video serve as secondary validation. All probes use angular or translational targets only.

(M4) Thumb R^2 is near zero for all models (best: 0.195) due to low target variance ($\text{std}=4.91^\circ$), and BIWI roll remains weak ($R^2=0.168$ Ridge) without attention pooling. These failure cases indicate that frozen probes struggle with low-variance targets and spatially diffuse signals. (M5) The CKA analysis (8 models, 28 pairs) yields $\rho=0.03$ ($p=0.88$), consistent with no relationship but not definitive proof of independence. With $n=28$ non-independent pairs (each model appears in 7 pairs), the minimum detectable ρ at 80% power is ≈ 0.50 ; moderate correlations (0.3–0.4) would go undetected. A Mantel test would better account for the shared-model dependence structure. (M6) We apply Holm-Bonferroni correction within each analysis family (TOST, Friedman) but not across analysis types; overall Type I error may be inflated. (M7) LoRA training uses 2,000 images drawn from the 6,400-image training split; we confirm no overlap with the 1,600-image probe test set.

Ethics and societal impact. FreiHAND and BIWI contain hand/face images collected with informed consent. Our probes extract aggregate joint angles, not identity-linked features; however, the same frozen-probing methodology could in principle be applied to surveillance-relevant tasks.

7 Conclusion

A single frozen backbone linearly encodes hand pose (6.1° MAE), head pose, object pose, and camera intrinsics, with each task adding only $\sim 6,000$ parameters. The text bottleneck reflects a pathway-training deficit—not a representational deficit—that LoRA partially recovers. Training objective determines accuracy more than architecture: a controlled ablation isolates the 0.15 gap between supervised and self-supervised/contrastive models. Five architecturally diverse encoders converge to equivalent accuracy ($R^2 \approx 0.55$) despite sharing as little as $\text{CKA}=0.41$ representational similarity, demonstrating functional convergence despite representational dissimilarity—extending the platonic representation hypothesis [49]. Spatial concentration provides an ablation-based explanation for cross-dataset variation. These findings suggest that frozen probing is both a scientific tool for understanding geometric representations and a practical approach to multi-task geometric measurement. Code and pre-trained probes will be released open-source.

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Appendix

A Full Per-Finger Results

Table 4 reports per-finger R^2 and MAE for all fourteen models on FreiHAND (8,000 images, RRR probe). The “uniform mean” column is the average across five fingers, treating each finger equally regardless of target variance. Thumb R^2 is near zero for all models due to low target variance (std = 4.91° vs. 13.0–16.2° for other fingers).

Table 4: Per-finger R^2 on FreiHAND (8,000 images). Best overall in **bold**.

Model	Thumb	Index	Middle	Ring	Pinky	Mean	MAE (°)
SigLIP 2 ViT-L	0.195	0.580	0.693	0.701	0.625	0.559	6.14
DINOv3 ViT-L	0.149	0.590	0.709	0.708	0.624	0.556	6.11
CLIP ViT-L	0.177	0.575	0.686	0.699	0.620	0.551	6.19
SigLIP ViT-L	0.175	0.569	0.688	0.706	0.614	0.550	6.16
InternViT-300M	0.174	0.571	0.681	0.699	0.607	0.547	6.12
DINOv2 ViT-L	0.143	0.551	0.667	0.662	0.594	0.523	6.39
Gemma 3 L0	0.140	0.547	0.638	0.644	0.556	0.505	6.59
DINOv2 ViT-B	0.117	0.523	0.627	0.602	0.540	0.482	6.75
Qwen-7B L8	0.138	0.531	0.621	0.599	0.511	0.480	6.83
SigLIP ViT-B	0.123	0.530	0.612	0.610	0.522	0.479	6.80
ConvNeXt-L S2	0.100	0.495	0.581	0.581	0.515	0.455	7.05
QwenViT L24	0.133	0.508	0.583	0.563	0.483	0.454	7.08
Qwen-3B L11	0.128	0.491	0.554	0.538	0.466	0.435	7.28
QwenViT-merger	0.105	0.482	0.546	0.534	0.459	0.425	7.32

B Controlled Architecture Ablation: Full Results

Table 5 reports the full per-finger results for the controlled ablation experiment (Sec. 4.5 of the main paper). DeiT3-L (ViT, ImageNet-1K) and ConvNeXt-L (CNN, ImageNet-1K) are matched on pretraining data; ConvNeXt-L (IN-22K+1K) shows the effect of scaling pretraining data alone.

Table 5: Controlled architecture ablation: per-finger R^2 on FreiHAND.

Model	Type	Thumb	Index	Middle	Ring	Pinky	Mean R^2	MAE (°)
DeiT3-L (IN-1K)	ViT	0.047	0.421	0.512	0.495	0.420	0.379	7.65
ConvNeXt-L (IN-1K)	CNN	0.102	0.451	0.516	0.502	0.455	0.405	7.49
ConvNeXt-L (IN-22K)	CNN	0.100	0.495	0.581	0.581	0.515	0.455	7.05

C BIWI Head Pose: Per-Component Results

Table 6 reports per-component R^2 on BIWI for Ridge and AttentionPool probes. Pitch is consistently easiest; roll is hardest for all models. AttentionPool substantially improves all components.

Table 6: BIWI head pose R^2 by component (Ridge / AttentionPool).

Model	Ridge				AttentionPool			
	Yaw	Pitch	Roll	Mean	Yaw	Pitch	Roll	Mean
DINOv3	0.705	0.948	0.168	0.607	—	—	—	0.838
DINOv2	0.668	0.874	0.052	0.532	0.958	0.940	0.779	0.892
SigLIP 2	0.385	0.902	0.078	0.455	—	—	—	0.787

D Per-Bone Joint Analysis

The proximal-distal gradient is quantified at the individual joint level. Table 7 reports Ridge and AttentionPool R^2 for all 15 joints grouped by position (5 joints per group). Proximal joints (MCP, PIP) far outperform distal joints (DIP).

Table 7: Per-bone R^2 for DINOv3 on FreiHAND (Ridge / AttentionPool).

Bone Type	Ridge R^2	AttnPool R^2	Count
MCP (proximal)	0.544	0.602	5
PIP (middle)	0.559	0.616	5
DIP (distal)	0.271	0.312	5
Mean (15 joints)	0.458	0.510	15

E Camera Intrinsic: Full Results

Table 8 reports Ridge R^2 for camera focal length (f_x) probing on FreiHAND. All models achieve high R^2 (0.81–0.94), with vision-only encoders leading. Autoregressive LLM processing reduces intrinsics prediction by 13.4%.

Table 8: Camera intrinsics probing (R^2 for f_x on FreiHAND).

Model	$R^2(f_x)$	Type
QwenViT	0.939	Vision encoder
DINOv3	0.924	Self-supervised
DINOv2	0.913	Self-supervised
SigLIP 2	0.902	Hybrid VL
Gemma 3 L0	0.887	Generative VLM
InternViT	0.883	Hybrid VL
CLIP	0.876	Contrastive VL
SigLIP	0.869	Contrastive VL
QwenViT-merger	0.829	Vision encoder
Qwen-7B	0.826	Generative VLM
Qwen-3B	0.813	Generative VLM

F DINOv2 Register Analysis

Table 9 compares DINOv2 (no registers) against DINOv2+registers at each probed layer, showing that registers accelerate early-layer geometric emergence but converge at optimal depth.

Table 9: DINOv2 vs. DINOv2+registers: layer-wise R^2 on FreiHAND.

Layer	DINOv2	DINOv2+reg	Δ
L4	0.114	0.207	+0.093
L8	0.160	0.269	+0.110
L12	0.300	0.368	+0.067
L16	0.432	0.441	+0.010
L20	0.523	0.541	+0.019
L23	0.510	0.522	+0.012

G Nested Cross-Validation Results

Table 10 reports nested 10-fold CV R^2 for the top models. Cluster models show test–CV gaps within 0.006, confirming that test-set hyperparameter selection introduces minimal bias for within-cluster comparisons. DINOv2 shows a larger gap (−0.029), consistent with its outlier status.

Table 10: Nested 10-fold CV vs. test-set R^2 on FreiHAND.

Model	Test R^2	CV R^2	Δ
SigLIP 2	0.559	0.563	+0.004
DINOv3	0.556	0.550	−0.006
CLIP	0.551	0.554	+0.003
SigLIP	0.550	0.549	−0.001
InternViT	0.547	0.549	+0.002
DINOv2	0.523	0.494	−0.029

H Patch Ablation Details

Table 11 reports the full patch ablation results for DINOv3 on BIWI and YCB-Video. Removing the highest-norm 100 patches has a large effect on BIWI (loosely framed) but minimal effect on YCB-Video (tightly cropped).

Table 11: Patch ablation: R^2 change from removing 100 patches (DINOv3).

Ablation	BIWI ΔR^2	BIWI (post)	YCB ΔR^2	YCB (post)
Top-norm patches	−0.126	0.481	−0.003	0.706
Random patches	−0.107	0.500	−0.040	0.669

I CKA Similarity Matrix

Table 12 reports linear CKA similarity between eight models (six ViT-L plus DINOv2-B and SigLIP-B) at their best layers on FreiHAND features (8,000 images). DINOv2 and DINOv3 share high representational similarity (0.881) yet achieve non-equivalent probing accuracy. SigLIP 2 and CLIP have the lowest CKA (0.412) yet achieve equivalent R^2 . Across all 28 pairs, Spearman $\rho = 0.03$ ($p = 0.88$) between CKA and $|\Delta R^2|$, confirming functional convergence without representational convergence.

Table 12: Linear CKA similarity on FreiHAND (8 models at best layers). Six ViT-L models plus DINOv2-B (L12) and SigLIP-B (L12).

	DINOv2	DINOv3	SigLIP	SigLIP2	CLIP	InternViT	DINOv2-B	SigLIP-B
DINOv2	1.000	0.881	0.703	0.523	0.565	0.708	0.861	0.659
DINOv3		1.000	0.691	0.522	0.558	0.681	0.800	0.643
SigLIP			1.000	0.526	0.524	0.744	0.659	0.771
SigLIP2				1.000	0.412	0.631	0.427	0.493
CLIP					1.000	0.554	0.470	0.510
InternViT						1.000	0.594	0.736
DINOv2-B							1.000	0.589
SigLIP-B								1.000

J Layer Curves: Full Data

Table 13 reports R^2 at 7 layer depths for the six ViT-L models. Contrastive models (SigLIP, CLIP) achieve higher mid-layer R^2 while self-supervised models (DINOv2, DINOv3) concentrate geometric information in deeper layers.

K YCB-Video: Full Per-Component Results

Table 14 reports per-component R^2 on YCB-Video (rotation: yaw, pitch, roll; translation: t_x , t_y , t_z). All models achieve similar rotation R^2 (≈ 0.70), with translation slightly lower. The task-dependent autoregressive degradation observed on hands dissolves on rigid objects.

Table 13: Layer-wise R^2 on FreiHAND for 6 ViT-L models.

Layer	DINOv2	DINOv3	SigLIP	SigLIP 2	CLIP	InternViT
L0	—	0.114	0.149	0.135	0.128	0.103
L4	0.114	0.204	0.276	0.265	0.238	0.246
L8	0.160	0.262	0.381	0.365	0.315	0.331
L12	0.302	0.340	0.519	0.507	0.435	0.432
L16	0.434	0.425	0.550	0.559	0.541	0.536
L20	0.523	0.556	0.549	0.557	0.551	0.547
L23	0.510	0.531	0.546	0.554	0.535	0.532

Table 14: YCB-Video per-component R^2 (Ridge probe).

Model	Yaw	Pitch	Roll	t_x	t_y	t_z	Rot	Trans
DINOv2	0.828	0.639	0.716	0.609	0.678	0.887	0.728	0.725
DINOv3	0.806	0.637	0.683	0.607	0.653	0.908	0.709	0.723
SigLIP 2	0.781	0.617	0.711	0.580	0.680	0.871	0.703	0.710
SigLIP	0.778	0.626	0.683	0.585	0.680	0.879	0.696	0.715
Qwen-7B	0.810	0.616	0.683	0.572	0.673	0.887	0.703	0.711

L Attention Head Analysis (DINOv2-L)

We probe each of the 16 attention heads in DINOv2-L layer 20 individually (Ridge regression on per-head output, 6,400 train / 1,600 test). The top-10 heads by R^2 are shown in Table 15. All heads achieve comparable geometric accuracy ($R^2=0.46\text{--}0.48$), with no evidence of joint specialization: the maximum absolute Spearman correlation between any head’s attention entropy and any single joint angle is $|\rho|=0.28$ (head 11, middle PIP). Geometry is an ensemble property distributed across all attention heads.

Table 15: Per-head probing R^2 for DINOv2-L layer 20 (top 10 of 16 heads).

Head	R^2	max $ \rho $	Best joint
6	0.479	0.167	middle PIP
14	0.478	0.035	ring MCP
4	0.475	0.273	index PIP
1	0.473	0.093	index MCP
11	0.473	0.280	middle PIP
12	0.469	0.042	ring MCP
8	0.466	0.160	middle PIP
9	0.463	0.056	middle PIP
2	0.460	0.076	ring PIP
10	0.459	0.090	index PIP

M Validity Controls

To confirm that probing results reflect genuine geometric encoding rather than dataset artifacts, we run three validity controls:

1. **Shuffled targets:** Randomly permuting labels across samples yields deeply negative R^2 for all models, ruling out spurious correlations between features and targets.
2. **Random features:** Gaussian noise features of matched dimensionality yield deeply negative R^2 , confirming the probe requires structured representations.
3. **Pixel baseline:** Raw pixel features (resized to 224×224 , flattened) achieve deeply negative R^2 , confirming that learned representations add substantial value beyond low-level statistics.

N Statistical Test Details

TOST equivalence testing. We apply two one-sided t -tests (TOST) with equivalence margin $\Delta = 0.03 R^2$ and Holm–Bonferroni correction across all $\binom{5}{2} = 10$ pairwise comparisons within the top-5 models. All pairs achieve $p < 0.05$ after correction, confirming statistical equivalence.

Friedman rank test. A Friedman test across 11 models on 10 CV folds rejects the null hypothesis of equal performance ($\chi^2(10) = 94.3, p < 10^{-15}$). Nemenyi post-hoc tests confirm that the top-5 equivalence cluster (SigLIP 2, CLIP, DINOv3, SigLIP, InternViT) differs significantly from the autoregressive VLMs (Qwen-3B, QwenVIT-merger).

O Gaze Direction Probing (MPIIFaceGaze)

Table 16 reports probing results for gaze direction (yaw and pitch) on MPIIFaceGaze (45,000 images, 15 subjects, 80/20 random split). DINOv3 dominates with $R^2 = 0.787$, primarily due to superior pitch prediction ($R^2 = 0.719$ vs. 0.360 for DINOv2). Yaw is consistently easier across all models ($R^2 = 0.71$ – 0.86). Model rankings differ from FreiHAND (hands), where the top-5 cluster shows no significant differences.

Table 16: Gaze probing on MPIIFaceGaze (45,000 images, RRR probe).

Model	R^2 yaw	R^2 pitch	R^2 mean	MAE ($^\circ$)
DINOv3	0.855	0.719	0.787	3.14
DINOv2	0.803	0.360	0.582	4.53
CLIP	0.854	0.259	0.557	4.70
SigLIP 2	0.844	0.248	0.546	4.74
ConvNeXt-L	0.709	0.249	0.479	5.11

P LoRA Layer Trajectory

Table 17 reports layer-wise probing results for Gemma 3 4B with and without the LoRA adapter ($r = 16, \alpha = 32$). Features are extracted at 10 layers across the 34-layer LLM and probed with RRR (rank sweep, 8,000 FreiHAND images). The LoRA delta grows from +0.025 at L0 to +0.117 at L28, showing that LoRA’s primary effect is preserving geometry at deep layers where the frozen base loses it.

Table 17: Layer-wise R^2 for Gemma 3 4B (LoRA vs. frozen base).

Layer	LoRA R^2	Frozen R^2	Δ
L0	0.530	0.505	+0.025
L2	0.532	0.504	+0.028
L4	0.524	0.492	+0.032
L8	0.529	0.483	+0.046
L12	0.491	0.438	+0.052
L16	0.403	0.361	+0.041
L20	0.365	0.307	+0.058
L24	0.332	0.236	+0.096
L28	0.314	0.197	+0.117
L33	0.283	0.181	+0.102